Recommended Assessment

Inverse Kinematics

Teach Pendant

1. Initially, the experiment was run using four end-effector positions to map the workspace. Now consider increasing the number of points to eight by adding additional waypoints along the edges of the rectangular path. How do you think this would affect the trajectory? Would using more points result in smoother motion between waypoints, or would there still be noticeable abrupt changes?

2. How accurately did the end-effector reach each recorded waypoint? Did you notice any deviations from the expected path? If so, what could be the possible causes?

3. What are some potential applications of using a teach pendant in an industrial environment, and what advantages does it offer?

4. What are the limitations of using teach pendant learning compared to pre-programmed motion paths in an industrial setting?

Trajectory Generation

5. How does the motion between waypoints get affected when you increase the duration from 3s to 6s? What trade-offs exist when increasing this value? Attach screenshots of your scope.

6. How does the motion between waypoints get affected when you decrease the duration from 3s to 6s? What trade-offs exist when decreasing this value?

7. Compare the trajectory generated in this procedure and the trajectory when doing Teach Pendant Follow. What differences do you observe, and why might they occur?